PART 03 · CHAPTER 10

Moving averages

Understand the smoothing before you trust the signal.

Answer edition: includes continuation, worked solutions and the knowledge-check answer.

intermediate About 5 min of prose · practice takes additional time AI editorial review 2026-10-12 · v0.4

▼ Review scope

Re-accepted for documented whitespace or shared study locator/date correction; earlier substantive review retained.

Recorded by /root. The review covers editorial evidence; it does not establish learning by human readers.

> THE IDEA TO KEEP
>
> An average compresses past prices into a smoother description; its usefulness as a trading rule is a separate, testable question.

By the end of this chapter

- Calculate a full-window SMA and SMA-seeded EMA, including undefined early values.
- Identify the timestamp of a close-based crossover.
- Contrast trend and range behavior under one frozen definition.

Builds on: Anatomy of a stock chart · Timeframes, sessions & chart scales · Trends, swings & changing structure

One line, one deliberate loss of detail

Imagine following a stock whose daily closes alternate between small advances and retreats. The individual marks can make its direction hard to describe. A moving average replaces each eligible observation with a summary of recent observations. You gain a simpler view, but you surrender information about their order and their extremes. Two quite different paths can have the same average. Begin by asking which detail you are willing to remove and why.

Fidelity defines a simple moving average, or SMA, as an arithmetic average over a specified period. Here the input is the closing price: SMA at time t equals the sum of the latest N closes divided by N. Five daily closes describe five sessions; five weekly closes describe five weeks. The parameter names a number of observations, rather than a universal duration.

Calculate it, then watch what leaves

For fictional closes of $10, $11, $12, $13, and $14, the five-session SMA is $12. If the next close is $15, the $10 observation leaves and $15 enters, producing $13. Notice that an average can move substantially because an old outlier leaves, even when today's change is small. An indicator has a memory structure; understanding that structure prevents you from attributing every movement in its line to something new in the market.

This textbook leaves the SMA undefined until all N observations exist. An implementation that averages only the observations available at the beginning produces a different early series. Adjusted versus unadjusted prices also change the input. Before comparing two charting systems, match the price field, session, adjustment policy, and missing-data handling. A smooth line is only as coherent as the data underneath it.

Main dataset: five-session averages

Session	Close ($)	SMA ($)	EMA ($)
1	10	Undefined	Undefined
2	11	Undefined	Undefined
3	12	Undefined	Undefined
4	13	Undefined	Undefined
5	14	12	12
6	15	13	13
7	15	13.8	13.667
8	14	14.2	13.778
9	13	14.2	13.519
10	12	13.8	13.012
11	11	13	12.342
12	10	12	11.561

Exponential weighting and the first value

An exponential moving average, or EMA, updates a prior average rather than dropping the oldest observation completely. A common convention is EMA today = alpha × close today + (1 − alpha) × EMA yesterday, with alpha = 2/(N + 1). For N = 5, alpha is one third. Fidelity describes using an SMA for the first EMA value. In our five-session example, seed the EMA at $12; the next $15 close produces $13.

On the following $15 close, the five-session SMA becomes $13.80, while the EMA becomes approximately $13.67. An EMA responds more to the newest price than the same-length SMA, but individual comparisons need not follow a simple ranking because the rest of the history matters. A seed's influence decays by repeated multiplication by 1 − alpha. Record the seed and warm-up history, especially in short samples.

The price of smoothing is delay

A longer lookback generally suppresses more local variation. It also carries older observations farther into the description of today. If a stock changes direction abruptly, those observations keep pulling the average toward the former regime. That is the source of lag. No choice eliminates both noise and delay. A very responsive setting exposes you to small fluctuations; a slower setting can remain persuasive after the underlying behavior has changed.

Suppose fictional price jumps from a long sequence near $20 to $24. An average below $24 is not an independent discovery that the stock is cheap. It is the arithmetic consequence of retaining earlier prices. Likewise, touching an average does not force a bounce. Use phrases such as "price returned to its recent mean" before adding a hypothesis about what participants might do there.

Observe the first state, then count crossings

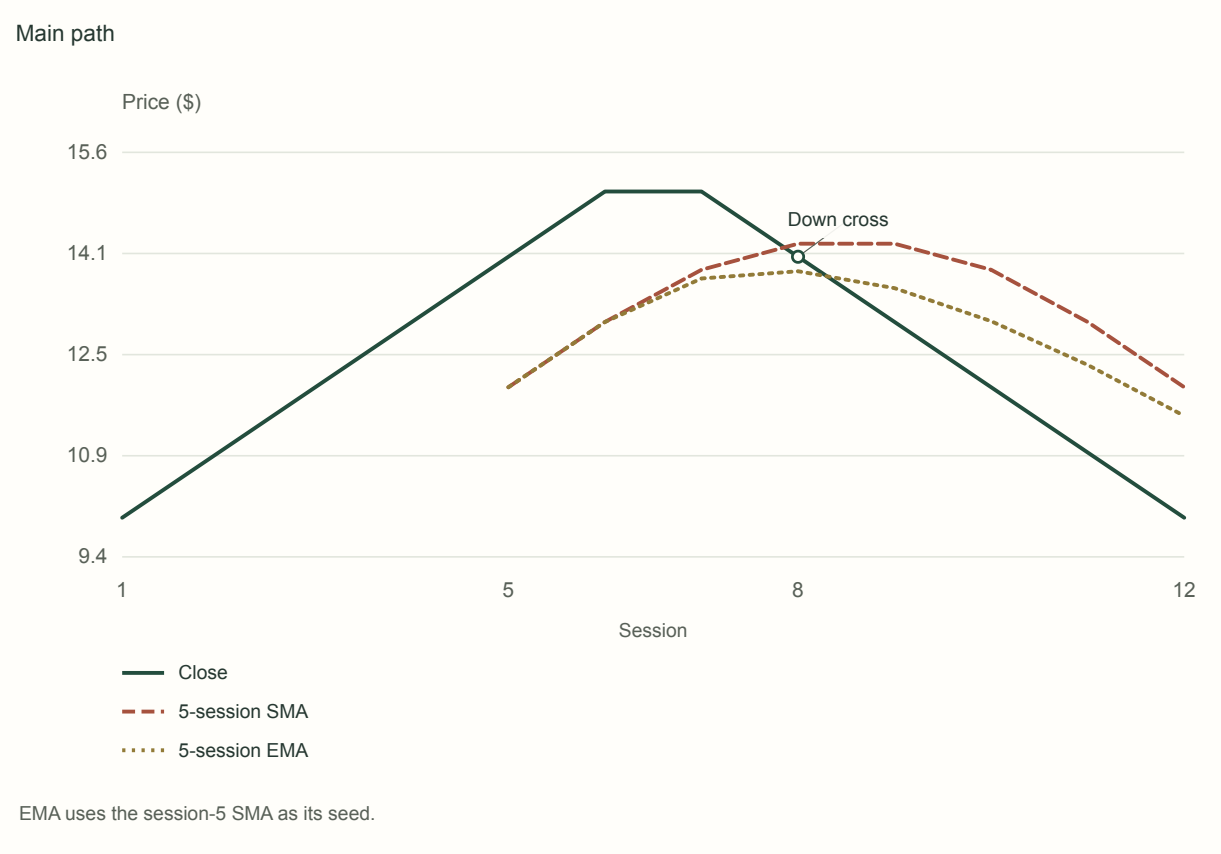
At session 5 of the main table, close $14 exceeds the newly available SMA $12. This is the first classified state, not a crossover: session 4 has no eligible mean. At session 7, $15 remains above $13.80. At session 8, $14 falls below $14.20 after the preceding close was above its own average. This first downward crossing is observable after session 8 closes.

Freeze this teaching definition: up means today's close exceeds its eligible SMA after yesterday was at or below its eligible SMA; down means today is at or below after yesterday was above. Both averages must exist. Equality belongs to the below-or-equal side. The definition identifies an analysis event, without assuming an execution at its closing price.

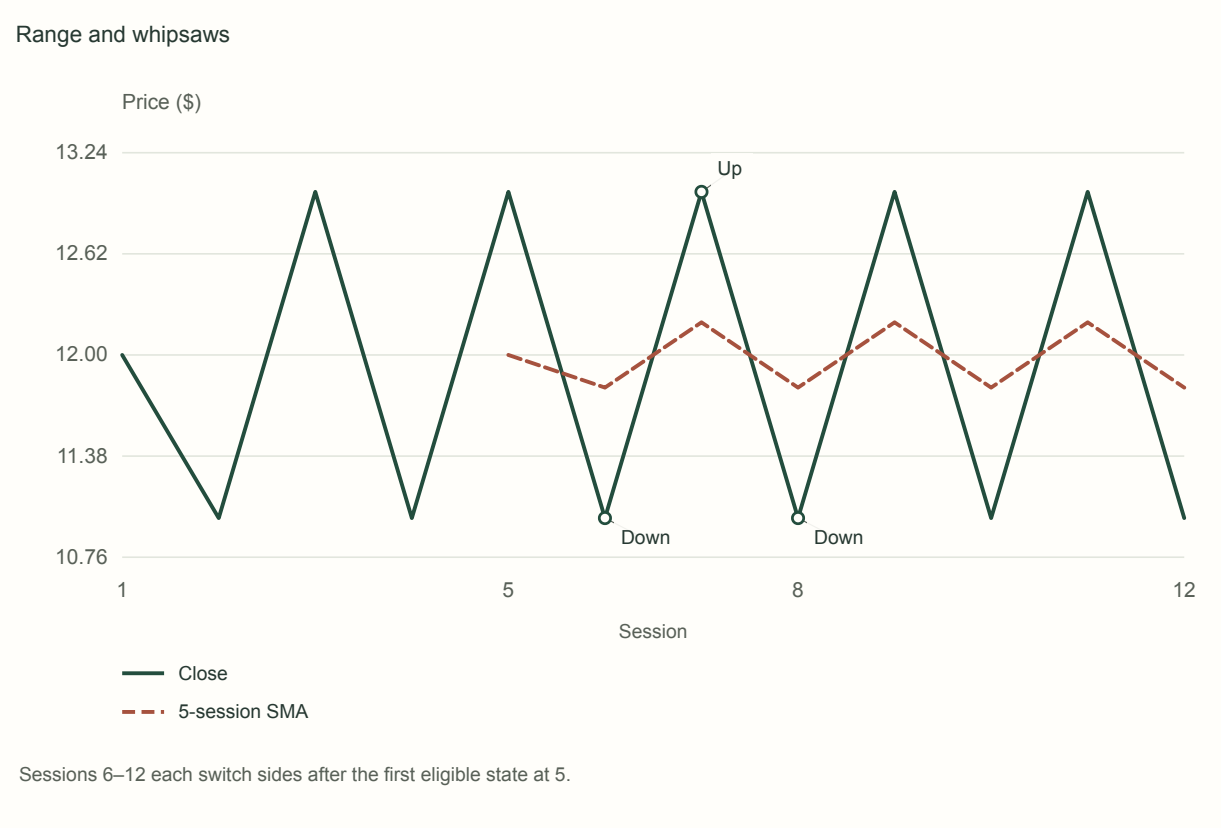
Apply the same definition to the contrasting range: $12, $11, $13, $11, $13, $11, $13, $11, $13, $11, $13, $11. Its first SMA is $12 at session 5, when price is above. From sessions 6–12 the close switches sides seven times. The definition has not changed; the path keeps undoing the previous directional event. The second panel makes those whipsaws visible.

FICTIONAL TEACHING ILLUSTRATION

One definition: a directional turn and a range



EMA uses the session-5 SMA as its seed.



Sessions 6–12 each switch sides after the first eligible state at 5.

Fictional 12-session datasets. Main values match the table, example, and exercise. The range is a deliberate contrasting path using the identical five-session SMA. Early averages are undefined.

▼ View the illustration's data

Main path

Session	Close	5-session SMA	5-session EMA
1	10	—	—
2	11	—	—
3	12	—	—
4	13	—	—
5	14	12	12
6	15	13	13
7	15	13.8	13.666666666666666
8	14	14.2	13.777777777777777
9	13	14.2	13.518518518518517
10	12	13.8	13.012345679012345

Session	Close	5-session SMA	5-session EMA
11	11	13	12.34156378600823
12	10	12	11.561042524005487

Range and whipsaws

Session	Close	5-session SMA
1	12	—
2	11	—
3	13	—
4	11	—
5	13	12
6	11	11.8
7	13	12.2
8	11	11.8
9	13	12.2
10	11	11.8
11	13	12.2
12	11	11.8

Read both panels before adding a filter

For the main path, write the observable sequence first: closes rise from $10 to $15, pause, and then decline. The mean turns after the closes turn, while price crosses it at session 8. The average supplies a fixed summary through the turn; it cannot identify the $15 peak before later prices exist. For the range, a more useful note is “closes alternate around a comparatively stable mean.”

Now require two consecutive closes on the new side before recognizing a change. In the range, none of the session-6-to-12 switches persists for two closes, so all seven are suppressed. In the main path, sessions 8 and 9 are the first pair below-or-equal, delaying recognition until 9. This is a concrete filtering tradeoff: fewer interruptions in this range, later response in this turn.

A reader could instead use a price buffer or a longer average, but those would be different definitions. Keep the demonstrated rule unchanged while comparing the panels. An annotation saying “down cross” means the ordering changed at that close; it does not show a completed trade or estimate its return. The planning and testing chapters will connect an observation to executable instructions and evaluation.

Two averages of the same closes describe overlapping information. With an equal five-session lookback, agreement shows that a reading survives a change of weighting, while disagreement reveals how differently the two lines weight and remember older closes. At session 8 here, the SMA is $14.20 and EMA about $13.78; the same $14 close is below one and above the other. Explain that arithmetic before calling either line a vote about the future.

A definition does not select a durable rule

Bajgrowicz and Scaillet’s DJIA experiment is relevant when a reader picks the best historical moving-average setting. Its adverse selection and cost findings concern a larger technical-rule family, not this exact SMA example. A crossover count is a path description; testing the rule still needs future evaluation and costs.

A WORKED EXAMPLE

Trace the session-8 crossing

Session 7’s SMA is (12 + 13 + 14 + 15 + 15)/5 = $13.80, below its $15 close. Session 8’s window is 13, 14, 15, 15, 14: $71/5 = $14.20, above its $14 close. The frozen rule identifies a downward event after session 8 closes. Its EMA is approximately $13.78, still below the close; the two weighting choices classify the same price differently.

Pause & try it

Calculate the main table’s session-7 and session-8 SMAs, identify the first downward crossing, and state when it becomes observable.

The five closes ending at 7 sum to $69, giving SMA $13.80. Those ending at 8 sum to $71, giving $14.20. Price goes from above its eligible mean ($15 > $13.80) to below ($14 < $14.20).

Record the crossing after session 8 closes. Session 5 merely establishes the initial state; comparing it with an undefined mean at 4 would manufacture a crossover.

GUIDED PRACTICE

Finish the EMA update

Use session-7 EMA $13.6666667, alpha 1/3, and session-8 close $14. Find the updated EMA and price's position relative to it.

EMA = 13.6666667 + (14 − 13.6666667)/3 = $13.7777778. Close $14 is above the EMA while below the SMA $14.20.

The indicators disagree because they retain different weights, not because one observed a different closing price.

INDEPENDENT PRACTICE

Compare a two-close filter

A fresh fictional sequence closes at $20, $22, $24, $21, $20, $23 and $25. Use a three-session SMA, initially undefined for sessions 1–2. Record above/below/equal states for sessions 3–7 and side switches only between eligible observations. A new side requires two consecutive strictly above or strictly below closes; equality resets the count. When are changes confirmed?

SMAs at sessions 3–7 are $22, $22.3333, $21.6667, $21.3333 and $22.6667. The close states are above, below, below, above, above. Session 3 establishes the initial state; side switches occur at 4 (down) and 6 (up).

The two-close rule confirms below at session 5 and above at session 7, each one close after the first crossing. Unlike the earlier alternating range, both changes persist in these supplied observations. The result is a property of this example, not evidence of an optimal filter.

CHECK YOUR UNDERSTANDING

One question before you go

What is the five-session SMA at main-dataset session 7?

A. $13.00

B. $13.67

C. $13.80

Answer: C. $69 divided by five gives $13.80.

Sources & further reading

- Fidelity · Simple moving average ↗
  Arithmetic moving-average definition and smoothing/lag tradeoff; no claim of profitable crossover rules. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- Fidelity · Exponential moving average ↗
  Recursive EMA definition and SMA seed convention. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- TradingView · Moving Averages ↗
  Conventional EMA smoothing multiplier 2/(N + 1). Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- Bajgrowicz & Scaillet · Technical trading revisited (author-hosted manuscript, sample to July 2011; published 2012) ↗
  Multiple-testing, ex ante persistence, transaction costs, and DJIA technical rules. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.

Teaching examples are fictional unless labelled historical. Inspect this chapter's evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

Trends, swings & changing structure ↗ Momentum: pace and persistence ↗

Evidence before conviction ↗

Practice & review this part →